

EBN111 SEMESTER TOETS/TEST 2 (8 MEI/MAY 2015)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:	MEMO	Studentennummer: Student number:							
Voorletters / Initials:		Taal / Language:	Afr		Eng				

Q1		Q2		Tot:	/45
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VRAAG 1 – QUESTION 1 (23)			
1.1	a) $V_o = 16V$ ✓	b) $V_o = 6V$ ✓	
1.2	a) $V_x = 2V$ ✓	b) $V_x = -8V$ ✓	c) $V_x = -6V$ ✓
1.3	$V_o = -10V$ ✓		
1.4	$R = 10\Omega$ ✓	$V = 10i_x + 10$ ✓	$i_x = 1A$ ✓
1.5	$R_{TH} = 30\Omega$ ✓	$V_{TH} = 0V$ ✓	
1.6	$I_L = 5/60 A$ or $0.833 A$ ✓		
1.7	a) $R_N = 8/3$ or 2.67Ω ✓	b) $I_N = 0.25A$ ✓	c) $I_L = 2A$ ✓

VRAAG 2 – QUESTION 2 (22)		
2.1	a) $R_I = 22\Omega$ ✓	b) $R_I = 10\Omega$ ✓
2.2	a) $v_o = -2V$ ✓	b) $v_o = -4V$ ✓
2.3	a) $v_o = 6V$ ✓	b) $i_o = -0.233 mA$ ✓
2.4	a) $v_x = -10.5V$ ✓	b) $V_{2min} = -4V$ ✓
2.5	$V_o = -2.5V$ ✓	
2.6	a) $V_I = 2mV$ ✓	b) $V_2 = 6V$ ✓